5. Jaccard Similarity

rA = {item 1, item 2, item 4, item 5}

rB = {item 1, item 3, item 4, item 5}

rA ∩ rB = {item 1, item 4, item 5}, |rA ∩ rB|=3

rA ∪ rB = {{item 1, item 2, item 3, item 4, item 5}, |rA ∪ rB|=5

Jaccard similarity (A,B) = |rA ∩ rB|/|rA ∪ rB| = 3/5 =0.6

Similarly,

Jaccard similarity of (A,C), (A,D), (A,E), (A,F), (B,C), (B,D), (B,E), (B,F) = 0.8,

Jaccard similarity of (C,D), (C,E), (C,F), (D,E), (D,F), (E,F) = 1.

Pearson Similarity

rowA=(8 + 1 + 2 + 7) / 4 = 4.5

rA={8 - 4.5, 1- 4.5, 0, 2 - 4.5, 7 - 4.5} = {3.5, -3.5, 0, -2.5, 2.5}

Similarly,

rB = {-2.75, 0, 0.25, 2.25, 0.25}

rC = { 1.8, -4.2, 1.8, -2.2, 2.8)

rD = { -0.4, -1.4, 1.6, -1.4, 1.6}

rE = {-3.6, 2.4, -0.6, 1.4, 0.4}

rF = {2.2, -2.8, 2.2, -2.8, 1.2}

Pearson similarity (A,B) = dot(rA,rB)/(|rA|*|rB|)

dot(rA,rB) = 3.5 * (-2.75) + (-3.5) * 0 + 0 * (0.25) + (-2.5) * 2.25 + 2.5 * 0.25

|rA| = [(3.5^2 + (-3.5)^2 + 0^2 + (-2.5)^2 + 2.5^2]^(1/2)

|rB| = [(-2.75)^2 + 0^2 + 0.25^2 + 2.25^2 + 0.25^2]^(1/2)]

Pearson similarity (A,B) = -0.6733

Similarly,

Pearson similarity of (A,C) = 0.9079; of (A,D) = 0.5962; of (A,E) = -0.8391; of (A,F) = 0.8733; of (B,C) = -0.4040; of (B,D) = -0.1154; of (B,E) = 0.7907; of (B,F) = -0.6221; of (C,D) = 0.8478; of (C,E) = -0.7017; of (C,F) = 0.9298; of (D,E) = -0.3007; of (D,F) = 0.7897; of (E,F) = -0.8139.

6. Rating of Item 2 by User B.

The Item-Item Collaborative Filtering should be used with $|N|=2$, where N is the number of nearest neighbors.

	User A	User B	User C	User D	User E	User F
Item 1	8	2	7	5	1	8
Item 2	1		1	4	7	3
Item 3		5	7	7	4	8
Item 4	2	7	3	4	6	3
Item 5	7	5	8	7	5	7

Similar to Exercise 8, here we consider items, rather than users.

Pearson similarity of (item 2, item 1) = Pearson similarity of (item 1, item 2) = -0.7864

Pearson similarity of (item 2, item 3) = -0.6013

Pearson similarity of (item 2, item 4) = 0.6663

Pearson similarity of (item 2, item 5) = -0.7186

Item 3 and item 4 have the top two Pearson similarity values with item 2.

Predict by taking weighted average: Rating of item 2 by user B is

(-0.6013*5 + 0.6663*7) / (-0.6013 + 0.6663) = 25.5015

Such a value of 25.5015 doesn't make any sense here, because one of the Pearson similarity values is minus. So normally in this case, if the Pearson correlation < 0, we don't use it for rating. Therefore, in the case of rating item 2 by user B, the answer will be 7, by only considering item 4.

Rating of Item 3 by User A.

Pearson similarity of (item 3, item 1) = Pearson similarity of (item 1, item 3) = 0.8626

Pearson similarity of (item 3, item 2) = -0.6013

Pearson similarity of (item 3, item 4) = -0.7432

Pearson similarity of (item 3, item 5) = 0.8444

Item 1 and item 5 have the top two Pearson similarity values with item 3.

Predict by taking weighted average: Rating of item 3 by user A is

(0.8626*8 + 0.8444*7) / (0.8626+ 0.8444) = 7.5053